Transfer task: defend and revise

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Support your prediction with measurements and calculations. Show a specific change and explain why you made it.

SUCCESS CRITERIA - I can...

1. Graph the central prediction and coordinate bounds.
2. Use calculations to support an answer. Explain when the calculation stops working.
3. Show a specific change and use measurements or details to explain why you made it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “revision” to explain your answer. Sentence starters are optional.

evidence: observations or measurements that support an idea

revision: a change made after checking evidence or reasoning

prediction: what a model says may happen next

01 Warm-up: bring back an earlier idea

For x_i=**3±1 m**, v=**5±0.5 m/s** and exact t=**7 s**, find the central final position and conservative interval. **Name a specific check of** your method.

Word help: conservative bound = a limit chosen to include every allowed case

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $x_f = x_i + v_x t$; $y_f = y_i + v_y t$.
Use your Day **21** data and consistent units. **Explain the endpoint** and what you treat as true. A bounded range is not a probability.

Worked reasoning: The cart calculation predicts **12 m** after **4 s**. The measured position is **10 ± 0.5 m**. Since **12 m** is outside **9.5–10.5 m**, check the calculation and what it treats as true. The difference alone does not tell you its cause.

Your turn: Mark the observed interval and prediction on a number line. **Name one thing the calculation** treats as true to recheck.

Sentence starter: I predict … . I used … to make this prediction because … .



Position (m); right is positive

Marzano check - circle: ① ② ③

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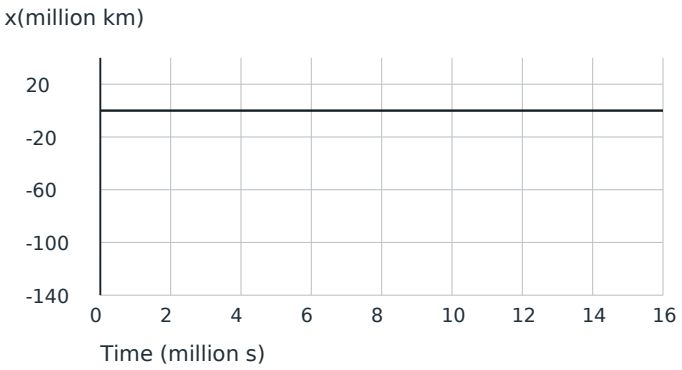
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Graph central and bounded positions

Use Day **21** Honors data. Graph central, lower and upper x(t) from **0** to **16** million seconds. Use additional graph paper if needed. **Mark end point values** and **explain the changing width**.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Marzano check - circle: 1 2 3

04 Support your answer for the public

Write an accessible paragraph with central end point, coordinate intervals, and the meaning of those bounds. Distinguish measurement uncertainty from constant-velocity calculation error.

Sentence starter: The step I would change is ... . I would change it because I found ... .

Explain to your audience	Evidence from your work
Predicted position	___
Range and meaning	___
Assumption / omitted effect	___

Marzano check - circle: 1 2 3

05 Peer defense and revision

Explain your method to a partner. Have them ask about units, axes, measurement limits, or something your calculation treats as true. **Record one question** and one change based on your measurements or observations. Keep the original visible. If working alone, label this a self-review.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Step checked	Question or concern	Change or reason to keep
___	___	___

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A **5 kg** object hangs still from a vertical string. Use g=**9.8 N/kg**. **Name all forces** on it with sources, sizes and directions. Does object-on-string belong on this diagram? **Draw the forces** on the chosen object and **label their sources**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

1. Point to finished work that shows each rubric category: **Science, Reasoning, Communication, and Transfer to asteroid**. **Explain how it meets the** category.
2. **Name your most important change** and one question for the next unit.

Sentence starter: I notice … . This supports … because … .

Criterion	Where your work shows it
Science / model	____
Reasoning	____
Communication / transfer	____

Marzano check - circle: ① ② ③

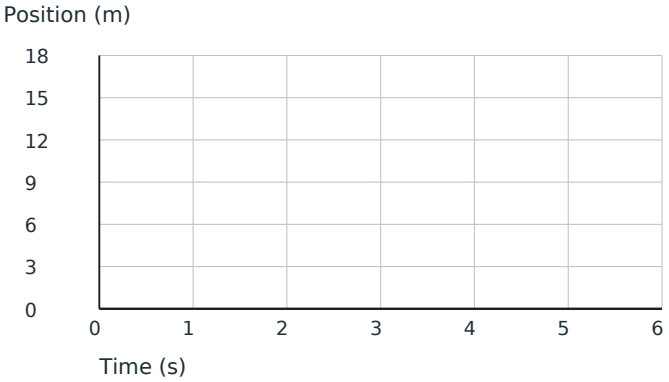
08 STRETCH: test the prediction

A class uses x = **10 m** at **4 s** to **find v = 2.5 m/s**. It then measures x = **13.0 ± 0.2 m** at **6 s**.

1. **Explain whether** the new measurement supports the constant-velocity prediction.
2. Propose a motion with **two time intervals** that fits the central measured positions.

Word help: evaluate = check a claim against evidence and conditions

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: ① ② ③